

A Study on the Universal Canonical Form Problem

A Framework, Independence Phenomena, Morphism-Class Exchange, and Optimal Involutive Colour Interfaces

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Preprint, 6 October 2026

Abstract

This paper studies the question: *does there exist, for mathematical objects, a canonical form that is simultaneously representation-independent, complete, functorial, compositional, and effectively computable?* The paper does not solve this question. It makes four contributions of differing weight. First, it separates five requirements commonly conflated under the phrase “canonical form” and proves that they are logically independent, in particular that weak compositionality is a free consequence of completeness while strong compositionality is a joint attribute of a code and a construction family (Section 2). Second, it assembles, with full attribution to classical work, the reduction chain from finite relational structures to relative-prime graphs, endowing it with intrinsic recovery and construction interfaces (Sections 4–5). Third, it proves a polarity classification of naturality: the morphism class with respect to which a structural transformation transports is governed exactly by the polarity (coordinate permutation versus complement) of its action on the presentation, ranging from the full homomorphism category down to induced embeddings (Section 5). Fourth, within an explicitly bounded per-colour rooted-gadget architecture, it poses and solves tightly the optimization problem of minimal equal-size colour interfaces supporting involutive transport: the optimal order is $\Theta(\sqrt{\log_2 q})$, attained with rooted radius 2 and optimal leading constant 2 in the bipartite-payload architecture (Section 7). Scope and failure boundaries are stated alongside every result; a classification appendix attributes each statement as established here, known background, or open.

1 Introduction

Mathematical objects are rarely given in one way. A finite group arrives as a permutation group, a matrix group, or a presentation; a graph arrives labelled or unlabelled, by adjacency list or by invariant; a structure in one language arrives translated into another. The phrase *canonical form* names the hope that behind these presentations there is a unique essential form, that objects can be put into it once and for all, and that mathematics can then be done *on the forms* — equivalence decided by comparing forms, constructions performed by operating on forms.

This hope, stated without care, is either trivial or false. It is trivial if “canonical form” means the isomorphism class: send every object to its class and completeness is free. It is false if it demands a natural coordinate labelling of every object: symmetry forbids it, by an elementary argument reproduced below. Between these two poles lies a spectrum of precise requirements, and the purpose of this paper is to map it.

Question 1.1 (Universal canonical form, informal). Does there exist, for a class $\mathcal{C}$ of mathematical objects with equivalence $\cong$, a canonicalization that is simultaneously representation-independent, complete, functorial on morphisms, compositional under the constructions of $\mathcal{C}$, and effectively computable with controlled complexity?

What this paper does not do. It does not answer Question 1.1 in general. The objects studied here are finite relational structures and finite graphs. Nothing is claimed about infinite structures, higher categories, or foundations. The word “universal” in our title names the question, not an achieved scope.

What this paper does.

- (i) It separates five requirements (completeness, functoriality, symmetry preservation, compositionality, effectivity) and proves their independence: weak compositionality is free (Proposition 2.5); strong compositionality is a joint attribute of the pair (code, construction family) and fails for natural codes on natural constructions (Proposition 2.7); faithful encoding does not preserve constructions (Proposition 2.8). A complete-but-incomposable and a composable-but-incomplete code are exhibited side by side.
- (ii) It assembles the reduction chain from finite relational structures over fixed signatures to *relative-prime* simple graphs (both the graph and its complement connected), with linear overhead and intrinsic recovery at every layer (Proposition 4.9). The classical components of this chain (modular decomposition, incidence encodings, colour-elimination gadgets) are cited, not rebranded; what is added is the intrinsic-recovery interface calculus.
- (iii) It proves a polarity classification of naturality (Theorem 5.5): a transformation of the presentation by coordinate permutations and complement flips transports exactly the homomorphisms that reflect every flipped relation. Coordinate permutation (digraph reversal) is natural on all homomorphisms; a complement flip of a loopless binary relation forces injectivity, i.e. induced embeddings — recovering, with classical attribution, the exchange pattern observed earlier in this programme.
- (iv) It poses the minimal equal-size colour-interface problem: how small can a family of q pairwise non-isomorphic, equal-order, equal-size rooted gadgets be, if colours must be intrinsically recoverable and colour permutations transported exactly? Within the per-colour rooted-gadget architecture the answer is $\Theta(\sqrt{\log_2 q})$, attained at rooted radius 2 with leading constant 2 optimal in the bipartite-payload architecture (Theorem 7.4).

Provenance. This paper consolidates a staged research programme (Stages I–XXII, archived as separate working notes) after an external novelty audit. Every statement carries one of three tags in the appendix: [E] established in the present study; [K] known background, cited; [O] open. Where a classical result is reproved for self-containedness, the tag is [K] with an internal proof.

2 The five-layer framework

2.1 Three strengths of canonicalization

Fix a category $\mathcal{C}$ of mathematical objects and write $\mathcal{C}^\simeq$ for its underlying groupoid (objects and isomorphisms).

Definition 2.1 (Canonicalization, three strengths).

- (i) *Semantic canonicalization* is the quotient map $N_{\text{sem}}(X) = [X]_{\cong}$.
- (ii) *Representative canonicalization* is the choice of a skeleton $\mathcal{K} \subseteq \mathcal{C}$ and a map $X \mapsto \text{Can}(X) \in \mathcal{K}$ with $\text{Can}(X) = \text{Can}(Y) \iff X \cong Y$.

- (iii) *Natural-coordinate canonicalization* additionally equips each object X with an isomorphism $\lambda_X: X \rightarrow \text{Can}(X)$, natural in every isomorphism $f: X \rightarrow Y$: either $N(f) \circ \lambda_X = \lambda_Y \circ f$ for a functorial action on canonical objects, or $\lambda_Y \circ f = \lambda_X$ in the strict variant.

Proposition 2.2 (Symmetry obstruction to strict coordinates; elementary, **[K]-adjacent**). *If $\lambda_Y \circ f = \lambda_X$ holds for every isomorphism $f: X \rightarrow Y$, then $\text{Aut}(X) = \{\text{id}_X\}$ for every object X . In particular, on the category of finite sets, no natural-coordinate canonicalization in the strict sense exists as soon as some object has a nontrivial automorphism (e.g. the two-element set).*

Proof. Take $f = a \in \text{Aut}(X)$: $\lambda_X \circ a = \lambda_X$, and cancelling the isomorphism λ_X gives $a = \text{id}_X$. For the two-element set with the swap s , naturality would give $\lambda(0) = \lambda(s(0)) = \lambda(1)$, contradicting injectivity. $\square$

Remark 2.3 (What survives). Proposition 2.2 obstructs only the *strict* variant. Under categorical naturality $N(f) \circ \lambda_X = \lambda_Y \circ f$ the symmetry is not killed but transported: conjugation by λ_X identifies $\text{Aut}(X) \cong \text{Aut}(\text{Can}(X))$. This correction — the canonical object must carry the morphisms, not merely replace the object — is the first structural lesson of this programme and motivates the functoriality requirement (C2) below.

2.2 The five requirements

Definition 2.4 (The five requirements). For a class $\mathcal{C}$ with equivalence $\cong$, a coding $C: \mathcal{C} \rightarrow \mathcal{D}$ and a specified family Ω of constructions:

(C1) Completeness. $C(X) = C(Y) \iff X \cong Y$.

(C2) Functoriality. C extends to the groupoid, with all choices of transport agreeing up to natural isomorphism.

(C3) Symmetry preservation. There is an isomorphism $\text{Aut}(X) \cong \text{Aut}(C(X))$, automatic under (C2); cf. Remark 2.3.

(C4) Compositionality, three grades. For each $\Omega \in \Omega$ there is B_Ω with $C(\Omega(X, Y)) = B_\Omega(C(X), C(Y))$; *weak*: B_Ω any function; *code-level*: B_Ω reads only codes; *strong*: B_Ω is computable in $\text{poly}(|C(X)| + |C(Y)|)$ without invoking a global re-canonicalization of $\Omega(X, Y)$.

(C5) Effectivity. C itself is computable, with controlled complexity.

Proposition 2.5 (Weak compositionality is free; **[E]**). *Every complete, representation-independent code is weakly compositional for every construction: the assignment*

$$B_\Omega(C(X), C(Y)) := C(\Omega(X, Y))$$

is well defined. Hence “completeness versus compositionality” is a substantive question only from the code-level grade upward, and genuinely open only at the strong grade.

Proof. If $C(X) = C(X')$ and $C(Y) = C(Y')$ then $X \cong X'$, $Y \cong Y'$, hence $\Omega(X, Y) \cong \Omega(X', Y')$ for any isomorphism-invariant construction Ω , hence $C(\Omega(X, Y)) = C(\Omega(X', Y'))$. $\square$

The next proposition locates the content of (C4) at the strong grade, on the smallest non-trivial class: finite simple graphs, with Ω either disjoint union or complement.

Definition 2.6 (Two codes on finite graphs). For a finite graph G on n vertices, fix the lexicographic order on vertex pairs and let $w(G)$ be the upper-triangular bit string of an adjacency labelling. The *lexicographic code* is $C_{\text{lex}}(G) = \min_{\sigma \in S_n} w(\sigma G)$, which is complete **[K]** (standard canonization folklore). For connected G let $c(G) = C_{\text{lex}}(G)$ and define the *component code* $C_{\sqcup}(G) = \{c(K) : K \text{ a component of } G\}$ (a multiset).

Proposition 2.7 (Strong compositionality is a joint attribute; [E], components of the observation [K]).

- (a) $C_{\sqcup}$ is complete on finite graphs and strongly compositional for disjoint union: $C_{\sqcup}(G \sqcup H) = C_{\sqcup}(G) \uplus C_{\sqcup}(H)$, the merge of multisets being linear.
- (b) C_{lex} is complete but its canonical code does not commute with complement pointwise: for the three-vertex single-edge graph G , $C_{\text{lex}}(G) = 001$ and $C_{\text{lex}}(\overline{G}) = 011$, whereas the pointwise complement of 001 is 110.
- (c) The vertex-count code $C(G) = |V(G)|$ is strongly compositional for disjoint union ($C(G \sqcup H) = C(G) + C(H)$) and maximally incomplete.

Hence at the strong grade, completeness implies neither compositionality nor conversely; compositional behaviour is a joint attribute of the pair (code, construction family), not of the code alone.

Proof. (a) Components of $G \sqcup H$ are exactly the components of G and of H ; merging multisets of codes of total size n costs $O(n)$ after sorting, and no global re-canonicalization is invoked. Completeness of $C_{\sqcup}$ follows from completeness of c on connected graphs and the fact that an isomorphism must permute components preserving isomorphism types and multiplicities. (b) Direct computation over the six labellings: the single-edge graph on three vertices realizes the three strings 100, 010, 001 and the two-edge graph (complement) realizes 011, 101, 110; minima are 001 and 011. (c) Immediate. $\square$

Proposition 2.8 (Faithful encoding does not preserve constructions; [E]). Let Q be the relative-prime marker embedding of Definition 4.2, which reflects isomorphism ($Q(G) \cong Q(H) \iff G \cong H$) and outputs connected, co-connected graphs. Then for all G, H ,

$$Q(G \sqcup H) \not\cong Q(G) \sqcup Q(H),$$

since the left side is connected and the right side is disconnected. Construction-preservation is therefore not inherited from faithfulness: the encoding must be redesigned to carry a construction interface (Section 5).

Remark 2.9 (Independence table). Witnesses produced in this programme: completeness without (strong) compositionality — Proposition 2.7(b); strong compositionality without completeness — Proposition 2.7(c); faithfulness without construction preservation — Proposition 2.8; and, forward-referenced, the exchange of naturality classes for constructions — Theorem 5.5.

3 Classical background and attribution

This section fixes the classical ground on which Sections 4–7 stand. Nothing here is claimed as new; the purpose is to draw the boundary honestly, once, instead of scattering attributions.

Canonical codes and graph canonization [K]. Lexicographic-minimum adjacency codes and canonical-labelling algorithms are standard (canonization folklore; practical labellers such as nauty/traces implement refinements). Proposition 2.7 uses only the completeness of the lexicographic code, which is elementary.

Modular decomposition [K]. Gallai’s theorem [1]: every finite graph admits a unique modular decomposition tree whose internal nodes are series, parallel, and prime; cographs are exactly the degenerate case with no prime nodes, with their cotree calculus for $\sqcup, \vee, (\cdot)$ classical [4]. In this programme, the recursive trichotomy “ G disconnected / $\overline{G}$ disconnected / both connected” reproduces a coarser, truncated version of that decomposition: we call graphs with G and $\overline{G}$ both connected *relative-prime*; these are carried, not resolved, by the union/join grammar. That isomorphism difficulty concentrates on prime graphs is a known phenomenon in algorithmic graph theory (e.g. the hereditary-class isomorphism dichotomy line [6]).

Encoding reductions and GI-completeness [K]. Isomorphism of finite relational structures over a fixed finite signature is interreducible with graph isomorphism: the incidence-graph construction (element vertices, tuple vertices, position-indexed attachment) is standard and appears e.g. in work on the graph isomorphism problem [3]. Colour- and label-elimination by local markers is folklore of the GI-completeness tradition [7]; reductions transferring completeness by subdivision, complement, and attached markers are routine tools. The specific linear markers used in Section 4 are instances of these standard moves and are tagged [K]-adjacent; only their recovery calculus is claimed ([E]).

Strong homomorphisms [K]. A homomorphism that also reflects adjacency ($f(u)f(v)$ an edge implies uv an edge) is called *strong*; the reflection property alone is sometimes called *full*. Injective strong homomorphisms are exactly the induced embeddings. These notions and their basic calculus are classical [2]. The polarity classification of Section 5 recovers and generalizes the corresponding exchange fact within this vocabulary.

Signed and two-edge-coloured structures [K]. Structures (V, R^+, R^-) partitioning the off-diagonal pairs are complete two-edge-coloured (signed) graphs; their homomorphism theory is developed [5]. In particular, homomorphisms between such *complete* signed structures are automatically injective — an elementary observation used repeatedly below and attributed here once.

Reconstruction [K]. The Kelly–Ulam reconstruction conjecture (every graph on at least three vertices is determined by its vertex-deletion deck) remains open [8]; recognizable and reconstructible families form an active literature. Our deck results concern specially marked families and make no claim on the conjecture.

4 The reduction chain and isomorphism-invariant recovery

Throughout, graphs are finite and simple. The purpose of this section is a single proposition (Proposition 4.9): finite relational structures over a fixed finite signature reduce, by a linear functorial chain with recoverable layers, to graphs that are *relative-prime* in the following sense.

Definition 4.1 (Relative-prime graphs). A graph G is *relative-prime* if G and $\overline{G}$ are both connected. (This is the carrier level of Gallai’s modular decomposition truncated at prime nodes; see Section 3.)

4.1 The marker construction

Definition 4.2 (Marker construction Q ; [E]). For a graph G with vertex set C , $|C| = n$, form $Q(G)$ as follows: add a root vertex r_0 adjacent to every vertex of C ; attach two pendant leaves to each vertex of C ; attach three pendant leaves to r_0 ; complete r_0 into a 5-cycle by four new vertices. Then $|V(Q(G))| = n + 2n + 1 + 3 + 4 = 3n + 8$.

Lemma 4.3 (Q lands in the relative-prime class; [E]). *For every G , the graph $Q(G)$ and its complement are connected.*

Proof. Connectedness of $Q(G)$: every vertex of C is adjacent to r_0 ; every leaf is adjacent to its attachment vertex; every cycle vertex reaches r_0 along the cycle.

For the complement $\overline{Q(G)}$, let L_0 denote the three leaves at r_0 , L_C the $2n$ leaves at core vertices, and Z the four cycle vertices besides r_0 . A leaf is non-adjacent in $Q(G)$ only to its own attachment vertex; a core vertex v is adjacent in $Q(G)$ only to r_0 , its G -neighbours, and its two leaves; a cycle vertex z only to its two cycle neighbours. Consequently *every vertex except r_0 is adjacent in $\overline{Q(G)}$ to every vertex of L_0* , while r_0 is adjacent in $\overline{Q(G)}$ to the two cycle vertices at distance two from r_0 , each of which is adjacent to all of L_0 . Hence $\overline{Q(G)}$ is connected. $\square$

Lemma 4.4 (Isomorphism-invariant recovery; [E]). *For a graph X define the isomorphism-invariant*

$$\ell_X(x) = \#\{y \in N_X(x) : \deg(y) = 1\}.$$

In $X = Q(G)$: $\ell_X(v) = 2$ exactly on $C = V(G)$; $\ell_X(r_0) = 3$; and $\ell_X = 0$ on all leaves and cycle vertices. Hence the induced subgraph $\text{Core}(X) := X[\{x : \ell_X(x) = 2\}]$ equals G , and the core is recovered by a prescription invariant under all isomorphisms.

Proof. Degrees in $Q(G)$: a core vertex has degree $\deg_G(v) + 3 \geq 3$; r_0 has degree $n + 3 + 2 \geq 6$; cycle vertices have degree 2; leaves have degree 1. The degree-one neighbours of a core vertex are exactly its two leaves (r_0 and G -neighbours have degree ≥ 3 , cycle vertices are not adjacent to it); thus $\ell(v) = 2$. The degree-one neighbours of r_0 are its three leaves. No leaf or cycle vertex has a degree-one neighbour: the single neighbour of a leaf has degree ≥ 3 , and the neighbours of a cycle vertex have degree 2 or more. Since ℓ is preserved by every isomorphism, the set $\{x : \ell_X(x) = 2\}$ and the induced subgraph on it are invariant. $\square$

Lemma 4.5 (Faithfulness; [E]). *$G \cong H$ if and only if $Q(G) \cong Q(H)$.*

Proof. An isomorphism $G \rightarrow H$ extends by $r_0 \mapsto r_0$, leaves to corresponding leaves, cycle to cycle. Conversely, an isomorphism $Q(G) \rightarrow Q(H)$ preserves ℓ , hence maps $\{x : \ell = 2\}$ onto $\{x : \ell = 2\}$, and its restriction is an isomorphism of the induced subgraphs, which are G and H by Lemma 4.4. $\square$

Remark 4.6 (Deletion decks of marked families; [E], low weight). Among the vertex-deletion cards of $Q(G)$, the card $Q(G) - r_0$ is singled out isomorphism-invariantly: it is the unique card with exactly three isolated vertices (deleting a core vertex leaves exactly two; deleting a leaf or cycle vertex leaves none). On this card the prescription of Lemma 4.4 still recovers G . This is a reconstruction statement for the marked family $\{Q(G)\}$ only and makes no claim on the Kelly–Ulam conjecture (Section 3).

4.2 The chain

The two remaining links are classical and are stated with their standard attributions; only the recovery calculus is ours.

Definition 4.7 (Incidence graph; [K], folklore of the isomorphism literature). For a finite relational signature $\tau = \{R_1^{(k_1)}, \dots, R_s^{(k_s)}\}$ and a finite τ -structure A , the *incidence graph* $I_\tau(A)$ is the vertex-coloured graph with: element vertices (colour E), one per element of A ; tuple vertices (colour R_i), one per tuple in R_i^A ; position vertices (colour $P_{i,j}$, $1 \leq j \leq k_i$); edges joining each R_i -vertex to its k_i position vertices, and each position vertex to the element occupying that position. Then $|V(I_\tau(A))| = |A| + \sum_i (1 + k_i)|R_i^A|$ and $A \cong B \iff I_\tau(A) \cong_{\text{col}} I_\tau(B)$, a colour-preserving isomorphism of incidence graphs being exactly an isomorphism of structures.

Definition 4.8 (Colour elimination; $[\mathbf{K}]$ -folklore, instantiated here). Let $\mathcal{C}$ be a finite colour set and $L: \mathcal{C} \rightarrow \{2, 3, \dots\}$ injective. For a $\mathcal{C}$ -coloured graph H , the graph $U_L(H)$ replaces each vertex v of colour c by v together with two attachment vertices adjacent to v , each carrying $L(c)$ pendant leaves; colours are discarded. Then $H \cong_{\text{col}} K \iff U_L(H) \cong U_L(K)$, and recovery is isomorphism-invariant: the attachment vertices are exactly $\{x : \ell(x) \geq 2\}$, the original vertices are exactly the vertices with two such neighbours, and the colour of v is recovered as L^{-1} of the leaf count of either of its attachment vertices. The construction is linear in $|V(H)|$ for fixed $\mathcal{C}$.

Proposition 4.9 (The reduction chain; $[\mathbf{E}]$, assembled from $[\mathbf{K}]$ links). *For every finite relational signature τ there is a map*

$$A \mapsto R_\tau(A) := Q(U_L(I_\tau(A)))$$

from finite τ -structures to relative-prime simple graphs such that:

- (a) $A \cong B \iff R_\tau(A) \cong R_\tau(B)$;
- (b) $|V(R_\tau(A))| = O(|A| + \sum_i |R_i^A|)$ with constant depending only on τ ;
- (c) *the structure, its incidence colouring, and the intermediate graph are each recoverable from $R_\tau(A)$ by isomorphism-invariant prescriptions.*

Proof. Each link is faithful on isomorphism classes (Definitions 4.7 and 4.8, Lemma 4.5); (a) follows by composition, and (b) by multiplying the three linear bounds, using $|V(Q(X))| = 3|V(X)| + 8$. For (c), the recovery prescriptions compose: from $R_\tau(A)$ recover the core by Lemma 4.4; from the core recover the coloured incidence graph by Definition 4.8; from it recover A by Definition 4.7. Each prescription is invariant under isomorphisms, hence so is their composition. $\square$

5 Transported operations and the exchange theorem

5.1 Transported operations

By Proposition 4.9, isomorphism classes of τ -structures inject into those of relative-prime graphs. Operations on the source side therefore *transport*: one defines the operation on the image by recovering, operating, and re-embedding.

Definition 5.1 (Transported operations; $[\mathbf{E}]$). On the image class of Q (finite graphs), define $X \boxplus Y := Q(\text{Core}(X) \sqcup \text{Core}(Y))$, $X \boxtimes Y := Q(\text{Core}(X) \vee \text{Core}(Y))$, $j(X) := Q(\overline{\text{Core}(X)})$,

where $\vee$ is the join.

Lemma 5.2 (Transported calculus; identities $[\mathbf{K}]$ -classical, transport $[\mathbf{E}]$). *On isomorphism classes, $(\boxplus)$ and $(\boxtimes)$ are associative and commutative, j is an involution, and the de Morgan identities hold: $j(X \boxplus Y) = j(X) \boxtimes j(Y)$ and dually. These hold because the bijection $[G] \mapsto [Q(G)]$ (Lemma 4.5) transports the classical identities $\overline{G \sqcup H} = \overline{G} \vee \overline{H}$ of graphs, which are elementary.*

Lemma 5.3 (Functoriality on homomorphisms; $[\mathbf{E}]$). *Every graph homomorphism $f: G \rightarrow H$ extends to a homomorphism $Q_*(f): Q(G) \rightarrow Q(H)$ by $r_0 \mapsto r_0$, cycle to cycle, leaves of v to leaves of $f(v)$; the assignment is faithful, and*

$$Q_*(f \sqcup g) = Q_*(f) \boxplus Q_*(g), \quad Q_*(f \vee g) = Q_*(f) \boxtimes Q_*(g).$$

Proof. Adjacency is preserved on each part: core edges by hypothesis on f ; a leaf ℓ at v has the single neighbour v , whose image has leaves as images; $r_0 v$ maps to $r_0 f(v)$; the marker maps identically. Faithfulness is clear on core vertices. For the two identities, both sides are Q -images of graphs linked by the same underlying vertex map. $\square$

Note what Proposition 2.8 already showed: these transported operations cannot be the *ambient* operations ($Q(G \sqcup H)$ is connected, $Q(G) \sqcup Q(H)$ is not). Transported operations are a calculus on the image class, not a preservation property of the embedding.

5.2 The exchange theorem

The homomorphism classes on which a structural transformation transports are not chosen freely; they are determined by the transformation. The following classification, established in Stage XXII of the programme, makes the determination exact for all transformations acting on the presentation.

Definition 5.4 (Presentation transformation). Fix a finite relational signature τ . A *presentation transformation* is a family $T = ((\sigma_i, \epsilon_i))_i$ with σ_i a permutation of the k_i coordinates of R_i and $\epsilon_i \in \{+, -\}$. For a τ -structure A on V , $T(A)$ has the same underlying set and $R_i^{T(A)} = \sigma_i \cdot X_i(A)$ where $X_i(A) = R_i^A$ if $\epsilon_i = +$ and $X_i(A) = \overline{R_i^A}$ if $\epsilon_i = -$; $\sigma \cdot R$ permutes coordinates, and the complement is the full $V^{k_i} \setminus R_i^A$, or the off-diagonal $(V^2 \setminus \Delta) \setminus R$ for loopless binary relations (the simple-graph convention). Symbols with $\epsilon_i = -$ are *flipped*. A homomorphism $f: A \rightarrow B$ is *T-stable* if the same set map is a homomorphism $T(A) \rightarrow T(B)$.

Theorem 5.5 (Exchange: stability is reflection; [E], core observation [K]-adjacent). $f: A \rightarrow B$ is *T-stable* if and only if f reflects every flipped symbol: $f(x) \in R_i^B \Rightarrow x \in R_i^A$ for each flipped i .

Proof. Fix a symbol i ; write R, σ . Since σ is a bijection on tuples, $\sigma \cdot \overline{R} = \overline{\sigma \cdot R}$. If $\epsilon_i = +$: preserving $\sigma \cdot R^A$ is implied by preserving R^A , because f acts componentwise and hence commutes with coordinate permutations: $y = \sigma x \in \sigma \cdot R^A$ with $x \in R^A$ gives $f(y) = \sigma f(x) \in \sigma \cdot R^B$. If $\epsilon_i = -$: preserving $\sigma \cdot \overline{R^A}$ says $\sigma^{-1}y \notin R^A \Rightarrow \sigma^{-1}f(y) \notin R^B$, i.e. $x \notin R^A \Rightarrow f(x) \notin R^B$; this is the contrapositive of reflection. $\square$

Corollary 5.6 (Two extremes).

- (a) ([E]) If no symbol is flipped, every homomorphism is *T-stable*. In particular digraph reversal ($\sigma = (12)$, $\epsilon = +$) transports the entire homomorphism category.
- (b) ([K], classical) Under the off-diagonal convention, if some flipped symbol R is binary and loopless in A and B , every *T-stable* homomorphism is injective and strong on R ; for simple graphs with T the complementation, the *T-stable* maps are exactly the induced embeddings [2].

Proof. (a) is immediate from the theorem. (b): let $u \neq v$. If $(u, v) \in R^A$, preservation of the loopless R^B forces $f(u) \neq f(v)$; if $(u, v) \notin R^A$, then $(u, v) \in \overline{R^A}$ and stability forces $(f(u), f(v)) \in \overline{R^B}$, which is off-diagonal, so $f(u) \neq f(v)$. Strength holds by preservation plus reflection. $\square$

Remark 5.7 (Polarity, not size, governs the exchange; [E]). Reversal and complementation are both single transpositions acting on a binary presentation, yet their stable classes are the full homomorphism category and the induced embeddings respectively. Unary flips (complementing one colour predicate) force reflection of that predicate — rigidity of the colouring — without injectivity, arity one offering no off-diagonal to separate points.

Theorem 5.8 (Simultaneous transport; [E]). Let Γ be a finite group of presentation transformations acting on the colour set of the incidence presentation by permutations π_γ , and let $\hat{\Theta}_\gamma$ denote the operation on the ambient relative-prime graphs induced, via the recovery of Proposition 4.9(c), by replacing each rooted marker (Section 6) of colour c by that of colour $\pi_\gamma(c)$. Then $\hat{\Theta}_\gamma(R_\tau(A)) \cong R_\tau(\gamma A)$, vertex and edge counts are preserved exactly, $\hat{\Theta}_\gamma \hat{\Theta}_{\gamma'} = \hat{\Theta}_{\gamma\gamma'}$, and

for every γ -stable homomorphism f the underlying set map of the extension of Lemma 5.3 is a homomorphism between the transformed graphs — the same set map serving all $\gamma \in \Gamma$ at once.

Proof. Marker replacement acts only at the isomorphism-invariantly recovered sites; equal order and size of the marker family preserve the counts; substitution composes, giving the group law. The homomorphism condition reduces, as in Lemma 5.3, to the source-level condition, which is stability. $\square$

6 Signed presentations and transported involutions

Definition 6.1 (Signed presentation; [K]). For a graph G on V , the *signed presentation* is $S(G) = (V, R^+, R^-)$ with R^+ the off-diagonal edges and R^- the off-diagonal non-edges; these partition $V^2 \setminus \Delta$. This is a complete two-edge-coloured (signed) graph; complementation of G is the involution σ exchanging R^+ and R^- .

Proposition 6.2 (Diagonal obstruction; elementary, [K]). *Every homomorphism of signed presentations is injective, hence an induced embedding of the underlying graphs.*

Proof. For $u \neq v$ the pair lies in exactly one of $R^\pm$, and preservation maps it into the corresponding relation of the target, which is off-diagonal; hence $f(u) \neq f(v)$. Preservation of both signs is preservation and reflection of adjacency. $\square$

Definition 6.3 (Rooted marker families; [E]). Let $\mathcal{C}$ be a finite colour set. A *rooted marker family* is an injective assignment $c \mapsto D_c$ of a rooted simple graph to each colour, attached in the colour-elimination step (Definition 4.8) at the isomorphism-invariantly recognizable attachment sites. The family is *equal-size* if all D_c have the same number of vertices and of edges. Equal-size families realize colour permutations as substitutions preserving the ambient counts exactly; the involution σ of signed presentations is thereby transported, through Proposition 4.9, to an involution of uncoloured relative-prime graphs (Theorem 5.8).

The mathematical question about rooted marker families is then not existence (families are easily manufactured) but their minimal order. That question is posed and answered, within an explicit class of encodings, in Section 7.

7 Optimal involutive colour interfaces

This section poses and solves, within an explicit class of encodings, the sizing problem for rooted marker families (Definition 6.3): how small can a family realizing q colours with exact size-preserving substitution be?

Definition 7.1 (The sizing problem). Let $\mathcal{C}$ be a finite colour set, $q = |\mathcal{C}| \geq 2$. A *rooted marker family of order m* assigns to each $c \in \mathcal{C}$ a rooted simple graph D_c with $|V(D_c)| = m$ such that:

(G1) **Equal size.** $|E(D_c)|$ is the same for all c .

(G2) **Distinguishability.** $c \neq c'$ implies $D_c \not\cong_{\text{root}} D_{c'}$.

(G3) **Ambient compatibility.** D_c contains no pendant vertex adjacent to the root-side attachment site that could confuse the invariant ℓ of Lemma 4.4-type recovery; concretely, in the ambient construction of Definition 6.3 the marker components are recognizable at their sites and their rooted types are read off isomorphism-invariantly.

(G4) **Substitution.** Colour permutations π act by substituting $D_c \mapsto D_{\pi(c)}$; by (G1) vertex and edge counts of the ambient graph are preserved exactly, and substitution composes.

Let $m(q)$ be the minimal order of such a family.

Three constructions of decreasing order were built in the programme's stage notes ($2q+5$ by chorded cycles; $O((\log q)^2)$ by balanced rooted trees; $O(\log q)$ by binary slots on a spine path). We record only the two that matter asymptotically.

Construction 7.2 (Bipartite carrier; [E]). Let $s \geq 2$ and let $A = \{0, \dots, s-1\}$, $B = \{s, \dots, 2s-1\}$. Fix the matching $F_0 = \{\{i, s+i\} : 0 \leq i < s\}$ on the s^2 bipartite positions, and let $K = \lfloor \frac{s^2-s}{2} \rfloor$. For each K -element subset F of the remaining $s^2 - s$ positions, define D_F as: a spine path $x_0 x_1 \dots x_{2s-1}$ rooted at x_0 ; a triangle on fresh vertices attached at each x_i ; a payload vertex y_i adjacent to x_i for each i ; payload edges $F_0 \cup F$ among the y 's. Then $|V(D_F)| = 10s$ and $|E(D_F)| = 13s - 1 + K$, independent of F , and the family over any $q \leq \binom{s^2-s}{K}$ choices of F satisfies (G1)–(G4).

Verification. Counts. Spine $2s$ vertices and $2s-1$ edges; triangles $3 \cdot 2s$ vertices and $3 \cdot 2s$ edges plus $2s$ triangle–spine links; payload $2s$ vertices, $2s$ payload–spine links, s matching edges, K variable edges: in total $10s$ vertices and $13s - 1 + K$ edges, independent of F .

Recovery (G2), (G3). The payload graph is bipartite, triangles do not meet it, and no edge joins two spine vertices beyond the spine itself; hence the triangles of D_F are exactly the attached triangles. The vertices adjacent to a triangle but not on it are exactly the spine vertices; induced on them is the spine path, and the root fixes its direction. The sole remaining neighbour of x_i is y_i , so all payload vertices and the edge set $F_0 \cup F$ are recovered; deleting the fixed F_0 leaves F . Thus $F \neq F'$ implies $D_F \not\cong_{\text{root}} D_{F'}$, and the rooted type is computed by an isomorphism-invariant prescription. No vertex of D_F carries a pendant leaf, so the invariant ℓ meets no collision inside the marker.

Substitution (G4). Replacing F by a fixed-cardinality F' preserves both counts; substitutions compose with the group law of permutations of the family. $\square$

Construction 7.3 (Constant-radius refinement; [E]). Let $t_L = \lceil \log_2(s+1) \rceil$, $t_R = \lceil \log_2(s+2) \rceil$. For each fixed-cardinality subset S of the $s(s+1)$ positions between left payloads $a_1, \dots, a_s$ and right payloads $b_1, \dots, b_{s+1}$, let D_S be: a root r ; hubs L, R with r, L, R a triangle; L joined to every a_i , R to every b_j ; markers $\alpha_1, \dots, \alpha_{t_L}$ and $\beta_1, \dots, \beta_{t_R}$ joined to r , where α_k (resp. β_k) carries $3+k$ pendant leaves and is joined to a_i (resp. b_j) exactly when the k -th binary digit of i (resp. j) is 1; and the cross edges S . Then $|V(D_S)| = 2s + O(\log^2 s)$, all counts are independent of S , the rooted radius is 2, and the family over $q \leq \binom{s(s+1)}{\lfloor s(s+1)/2 \rfloor}$ choices of S satisfies (G1)–(G4).

Verification. The triangle rLR is unique: the payload is bipartite and the cross edges join only a 's to b 's, while L meets no b and R no a ; the markers carry no triangles. Within it, r is the vertex adjacent to no payload, and L is distinguished from R by payload counts s versus $s+1$. Payload labels are read off the marker adjacencies (markers are mutually distinguished by their pendant-leaf counts $3+k \geq 3$, which also keeps $\ell \neq 2$ off the attachment sites), and S is then read off the cross edges. Radius 2 at r : hubs and markers at distance 1, payloads and marker leaves at distance 2. Substitution acts on S as in Construction 7.2. $\square$

Theorem 7.4 (Optimal order of involutive colour interfaces; [E]).

(i) (Counting lower bound; holds already under (G2) alone.) *Every family of q pairwise rooted non-isomorphic graphs of order m satisfies*

$$q \leq m \cdot 2^{\binom{m}{2}}, \quad \text{hence} \quad m \geq (1 - o(1)) \sqrt{2 \log_2 q}.$$

(ii) (Upper bound.) *With s minimal such that $\binom{s^2-s}{\lfloor s(s+1)/2 \rfloor} \geq q$, Construction 7.2 gives a family of order $10s$; consequently $m(q) = \Theta(\sqrt{\log_2 q})$.*

(iii) (Constant radius and optimal leading constant.) *Construction 7.3* gives families of rooted radius 2 and order $2\sqrt{\log_2 q} + o(\sqrt{\log_2 q})$. Moreover, in the bipartite-carrier class — families whose colour information is carried by a fixed-cardinality edge set on the positions of a fixed bipartite subgraph, all other data common to the family — every family has order at least $2\sqrt{\log_2 q}$, so the leading constant 2 is optimal in that class.

Proof. (i) The number of isomorphism classes of rooted graphs on m vertices is at most the number of labelled rooted graphs, which is $m \cdot 2^{\binom{m}{2}}$: m choices of root and $2^{\binom{m}{2}}$ labelled graphs. An injective assignment of colours requires q distinct isomorphism classes. Taking logarithms, $\log_2 q \leq \log_2 m + \binom{m}{2}$; since $\log_2 m = o(\log_2 q)$ along any realizing sequence (otherwise $\binom{m}{2}$ could not absorb $\log_2 q$), we get $m(m-1) \geq 2\log_2 q(1 - o(1))$, i.e. $m \geq (1 - o(1))\sqrt{2\log_2 q}$.

(ii) By the central binomial estimate $\binom{N}{\lfloor N/2 \rfloor} \geq 2^N/(N+1)$ with $N = s^2 - s$, the minimal s satisfies $s^2 - s \geq \log_2 q - O(\log \log q)$, so $s \leq (1 + o(1))\sqrt{\log_2 q}$ and the order $10s$ is $O(\sqrt{\log_2 q})$; combined with (i) this is $\Theta(\sqrt{\log_2 q})$.

(iii) Existence: with $N = s(s+1)$, the same estimate gives $s = (1 + o(1))\sqrt{\log_2 q}$; the order $2s + O(\log^2 s) = 2\sqrt{\log_2 q} + O(\log^2 \log q) = (2 + o(1))\sqrt{\log_2 q}$, and the rooted radius is 2 by *Construction 7.3*. Lower bound in the class: let a bipartite-carrier family have bipartite subgraph with sides a, b , $n = a + b$ payload vertices. Colour information of fixed cardinality K on the ab positions yields

$$q \leq \binom{ab}{K} \leq 2^{ab} \leq 2^{n^2/4},$$

the last step by $ab \leq n^2/4$. Hence $n \geq 2\sqrt{\log_2 q}$, and the order $m \geq n \geq 2\sqrt{\log_2 q}$. $\square$

Remark 7.5 (Scope of Theorem 7.4). The lower bound (i) constrains any per-colour rooted-marker encoding; the lower bound in (iii) constrains only the bipartite-carrier class. Whether encodings outside both — global, non-per-colour schemes — can realize q colours with $o(\sqrt{\log_2 q})$ overhead is open (Section 8). Constant factors for radius-2 families outside the bipartite class are likewise open.

8 Scope, failure boundaries, and open problems

What is established. For finite relational structures over fixed finite signatures: the five-layer framework and its independence phenomena (Section 2); a linear reduction chain into relative-prime graphs with isomorphism-invariant recovery of every layer (Proposition 4.9); a transported-operation calculus and its faithfulness on homomorphisms (Section 5); an exact polarity classification of the morphism classes along which presentation transformations transport (Theorem 5.5); and the optimal order of equal-size involutive marker families within the stated classes (Theorem 7.4).

What is not established. Nothing is claimed about infinite structures, higher categories, or foundations: the universal question (Question 1.1) remains open in every direction beyond finite relational structures. The strong compositionality requirement (C4, strong grade) has positive instances (component codes under disjoint union; the transported calculus) and negative instances (lexicographic codes under complement; ambient preservation under disjoint union, Proposition 2.8), but no classification. Effectivity (C5) is untouched: our constructions are polynomial-time, but no complexity lower bounds for canonicalization itself are proved.

Open problems.

(O1) Naturality classes for transformations not of presentation type: products, quotients, and general functorial constructions (cf. Theorem 5.5).

- (O2) Colour encodings outside the per-colour rooted-marker class: can the $\Theta(\sqrt{\log_2 q})$ overhead of Theorem 7.4 be beaten, e.g. by global schemes, or is the counting obstruction universal?
- (O3) Constant factors for radius-2 marker families outside the bipartite-carrier class; minimal radius in general.
- (O4) Complement conventions for relations of arity ≥ 3 and their separation behaviour (cf. Corollary 5.6(b) and its convention dependence).
- (O5) Deletion-deck reconstructibility beyond marked families, and its precise relation to the Kelly–Ulam conjecture (Remark 4.6).
- (O6) The equivalence, if any, between strong compositionality of a complete code and the efficient computability of canonical labellings — the complexity-theoretic content of requirement (C4)+(C5).

9 Classification appendix

Every numbered statement of this paper is classified below. **[E]**: established in the present study. **[K]**: known background, cited; a proof included for self-containedness does not change a **[K]** tag. **[O]**: open, listed in Section 8. The classification follows an external novelty audit of the programme’s stage notes (Stages I–XXII), conducted by literature search; the audit is archived with the programme.

Section 2 (framework). Definitions 2.1 and 2.4 (strengths of canonicalization; five requirements with graded compositionality): definitional contributions, **[E]**. Proposition 2.2 (symmetry obstruction): elementary, **[K]**-adjacent, proof included. Proposition 2.5 (weak compositionality is free): **[E]** as a clarification. Proposition 2.7: the codes are **[K]** (standard canonization folklore); the juxtaposition and the strong-compositionality analysis are **[E]**. Proposition 2.8: **[E]** (uses Definition 4.2).

Section 3 (background). All **[K]**, with citations as given there.

Section 4 (reduction chain). Definition 4.1: **[K]**-context (Gallai truncation). Definitions 4.7 and 4.8: **[K]** and **[K]**-folklore; the instantiation and the recovery prescriptions are part of **[E]** below. Definition 4.2 and Lemmas 4.3, 4.4, 4.5: **[E]** (within the **[K]**-folklore genre of marker reductions). Remark 4.6: **[E]**, low weight, no claim on the Kelly–Ulam conjecture. Proposition 4.9: **[E]** as an assembly of **[K]** links with the **[E]** recovery calculus.

Section 5 (transported operations and exchange). Definition 5.1 and Lemma 5.2: transport **[E]**, identities **[K]**. Lemma 5.3: **[E]**. Definition 5.4 (presentation transformations): **[E]**. Theorem 5.5: **[E]**, with the core observation **[K]**-adjacent (the reflection calculus is classical). Corollary 5.6: (a) **[E]**; (b) **[K]**, classical, attributed. Theorem 5.8: **[E]**.

Section 6 (signed presentations). Definition 6.1 and Proposition 6.2: **[K]**. Definition 6.3 (rooted marker families): **[E]**, definitional.

Section 7 (sizing). Definition 7.1, Constructions 7.2 and 7.3, and Theorem 7.4: **[E]**; the counting technique of part (i) is standard, but the problem, the constructions, and the tightness statement are the contribution. The earlier constructions mentioned in passing ($2q + 5$; $O((\log q)^2)$; $O(\log q)$) are **[E]** and subsumed.

Section 8 (scope). The six open problems (O1)–(O6): [O].

Lineage. The stage notes map onto this paper as follows: Stages I–II → Sections 1–2; Stages III–XII → Sections 4–5 (transported calculus); Stages XIII–XV → Section 5 (exchange), with Stages XIV–XV’s observations re-attributed to the classical strong homomorphism literature by the audit; Stages XVI–XXI → Sections 6–7; Stage XXII → Theorem 5.5. Stage material judged [K] after the audit is cited rather than re-proved, except where a short proof aids self-containedness.

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